

Written Analysis

1: Heuristic design and admissibility

Steps:

$h(n) = x$ where x is the number of different letters from the current state to the goal state.

Since every action changes exactly one letter position, fixing x incorrect positions requires at least x actions. This heuristic's lower bound is the minimum number of actions needed to reach the goal.

For steps, the actions are consistent, with each step always costing the same amount. This means that just counting how many steps that are left can't overestimate the remaining cost.

Scrabble:

$h(n) = \sum \text{scrabbleValues}(\text{goal}[i])$ summed over all the positions where the current state differs from the goal state.

Similar to steps, every action changes exactly one letter position, but there are values applied to each letter. So, summing the values of all the letters needed would be the lower bound.

There is not an optimal path as the order of the letter changes does not affect the cost. So, this also means that if each step changes one of the current letters to match a letter of the goal state, we will reach the goal with the minimum cost.

Frequency:

$h(n) = x + \text{rarity}(\text{goal})$ where x is the number of different letters from the current state to the goal state, and $\text{rarity}(\text{goal})$ is the goal word's rarity value from the dictionary.

At least x actions are needed to fix x differences between current and goal state, and each action costs at least 1. The last action must produce the goal word exactly, so that one costs exactly $1 + \text{rarity}(\text{goal})$. Adding these together gives $x + \text{rarity}(\text{goal})$ as the minimum possible cost.

We know the last action's exact cost, and we know every other action costs at least 1. Since we never guess higher than what's actually guaranteed, the heuristic can't overestimate the true cost.

2: Does the heuristic actually help?

Word Pair	Cost Measure	USC Cost	USC Nodes Expanded	A* Cost	A* Nodes Expanded	Reduction
Why-Ask	steps	5	773	5	16	97.9%
Why-Ask	scrabble	9	765	9	13	98.3%
Why-Ask	frequency	38.86	808	38.86	755	6.6%
Bear-Duck	steps	4	691	4	6	99.1%
Bear-Duck	scrabble	11	2981	11	6	99.8%
Bear-Duck	frequency	38.24	1084	38.24	720	33.6%

1) Yes, both USC and A* always had the same optimal cost. Both algorithms' goals are to find the optimal solution, so this is expected.

2)

Why-Ask: Scrabble 98.3%

Bear-Duck: Scrabble 99.8%

3)

Why-Ask: Frequency 6.6%

Bear-Duck: Frequency 33.8%

4)

Steps and Scrabble's heuristics were very informative while frequency's heuristics was not very helpful. For steps and scrabble, the heuristics capture the lower bound, but also the true cost. For those methods, it is easier to predict the steps, while for frequency, the bound is looser. The heuristic only knows the true cost of the final action. So frequency basically only predicts the bare minimum.

Question 3: Reversibility and direction-dependent costs

Steps:

Yes, each step costs the exact same and the number of different letters would be the same in either direction.

Scrabble:

No, the number of steps needed would be the same, but the letters have different values so the steps would have different costs.

Frequency:

No, words have different rarity values so the values won't be the same. The in between steps may be the same, but the goal states will most likely have different rarities.

Question 4: LLM heuristic audit

Using ChatGPT 5.6 Luna

Prompt:

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic h_0 (n) is the number of positions in which state n differs from the goal.

Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal.

By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

Show less ^

Proposed Heuristic:

For a non-goal state n ,

$$h_R(n) = d(n, g) + r(g)$$

where:

- $d(n, g)$ is the Hamming distance between n and the goal g
- $r(g)$ is the rarity of the goal word.

For the goal,

$$h_R(g) = 0.$$

Precise Formula:

Let $d(n, g)$ be the number of positions where n and g differ.

$$h_R(n) = \begin{cases} 0 & n = g \\ d(n, g) + r(g) & n \neq g \end{cases}$$

Is it admissible?

Yes, this heuristic counts the steps needed, and it adds the rarity level for the goal state. The cost will be a minimum of 1 for each of the steps, but we don't know the intermediate states to get the rarity. However, we do have the goal state, so that can be used in the heuristic.

4) The proposal is valid; we cannot add possible rarities for the intermediate words because we could over predict the cost.

5)

Why-Ask: 6.6% reduction

Bear-Duck: 33.8% reduction

The LLM's frequency heuristic does not compare with the heuristics of steps and scrabble.